Yiling Xu

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Education

New York University Aug 2025 – May 2027
Master of Science in Data Science | GPA: 4.0/4.0 *New York, New York*

University of Washington Oct 2020 – Aug 2024
Bachelor of Science in Informatics, Data Science Track *Seattle, Washington*

Professional Experience

ByteDance | *Machine Learning Engineer Intern* Jul 2026 – Sep 2026

- Developed a multimodal LLM agent combining sampled video frames, ASR transcripts, titles, and metadata to classify livestreams across 20+ categories and 100+ fine-grained labels.
- Built a batch inference and sampling workflow with concurrent requests and majority voting to process livestream inputs efficiently while balancing inference latency, API cost, and prediction reliability.
- Improved offline classification accuracy from 71% to 94.6% on a benchmark of 2,600+ human-labeled livestream samples through hierarchical prompt design and iterative error analysis.
- Evaluated the LLM agent in an online A/B test on 2% of livestream traffic; observed neutral results on monitored metrics, complementing offline classification benchmarks.

Jutze Intelligence Technology | *Data Scientist Intern* May 2026 – Jul 2026

- Fine-tuned a BiSeNet semantic segmentation model for automotive surface defects, reducing the missed-defect rate by approximately 10 percentage points versus the original checkpoint.
- Built an image preprocessing and tiling workflow transforming 37 annotated high-resolution images into 6,924 training samples while preserving small defect boundaries.
- Evaluated performance with mIoU, class-wise IoU, precision, recall, and visual error analysis to investigate missed detections and false positives.

Tineco | *Machine Learning Engineer Intern* May 2025 – Jul 2025

- Fine-tuned a BERT-based multiclass NLP model to distinguish device faults, user intents, and false triggers in noisy real-world voice commands.
- Cleaned and structured 8,000+ voice-command records for model training and evaluation, addressing ambiguous text and inconsistent input quality.
- Developed a repeatable preprocessing, training, evaluation, and batch-inference workflow to support error analysis and model comparison.

Jiangsu EKX Intelligent Technology | *Data Scientist Intern* Jun 2023 – Sep 2023

- Developed AWS Redshift SQL pipelines to clean, join, reconcile, and validate 150K+ operational records, including anomaly and data-quality checks.
- Standardized KPI definitions in reusable SQL transformations and migrated Tableau reporting to Amazon QuickSight, improving metric consistency across teams.
- Published curated datasets and self-service dashboards for recurring operational analysis, reducing manual reporting preparation for stakeholders.

Data Science Projects

NYU VIP: Energy Commodity Markets | *Data Scientist* Jan 2026 – Present

- Compared XGBoost, LSTM, and Transformer forecasts on energy commodity time series to investigate nonlinear patterns and temporal dependencies.
- Built a reproducible Pandas, scikit-learn, and PyTorch framework for preprocessing, feature engineering, chronological splits, and model evaluation; ran weekly controlled experiments and ablations.
- Analyzed errors across forecast horizons, sequence lengths, and feature configurations to guide subsequent experiments.

COVID-19 Topic Modeling & Visualization | *Python, NLP, Shiny* May 2024 – Nov 2024

- Built an end-to-end text-analysis workflow using 100,000+ COVID-19 Twitter posts and public epidemiological data.
- Engineered TF-IDF and n-gram features and trained NMF topic models to extract interpretable themes in public discussions.
- Developed an interactive Shiny dashboard visualizing topic trends over time and across regions alongside case counts and outcome statistics.

Research Experience

NYU Immersive Computing Lab | *Research Assistant* Apr 2026 – Present

- Designed a benchmark covering AI-generated and real-world video distortions to evaluate artifact localization, correction quality, and robustness across failure modes.
- Prototyped selective video correction using salience scoring and masked inpainting to target perceptually important regions rather than regenerating entire videos.

Carnegie Mellon University | *Research Assistant* Feb 2023 – Dec 2023

- Built an automated evaluation framework to benchmark fully homomorphic encryption schemes across runtime, memory use, ciphertext expansion, accuracy, and security settings.
- Analyzed performance and security trade-offs across encryption methods and authored a first-author research paper published at ICDSE 2024.

Technical Skills

Programming & Analysis: Python, SQL, R, Pandas, NumPy, scikit-learn, Jupyter, MATLAB
Machine Learning: XGBoost, PyTorch, Transformers, BERT, NLP, Time-Series Forecasting, Classification, Model Evaluation
Data Platforms & Tools: AWS Redshift, MySQL, Tableau, Amazon QuickSight, Git, Linux, Weights & Biases